

Md. Rifat Aknda

Prospective Research-Based Master's Student (AI/ML)

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Research Interests

Artificial Intelligence & Machine Learning | Computer Vision | Multimodal Machine Learning (Signal, Image, Text) | Biomedical Signal Processing | Deep Learning & Edge AI | Transfer Learning and Feature Fusion | Wearable Computing & Human-Centered AI | Explainable AI for Clinical Decision Support

Research Output Summary

- Journal articles: 1
- Conference papers: 3 (IEEE)
- Best Paper Award: 1

Education

B.Sc. in Computer Science & Engineering | University of Liberal Arts Bangladesh (ULAB)
CGPA: 3.51 / 4.00 | Graduated: 2025

Publications

Journal Articles

- **M. R. Aknda**, F. A. Farid, J. Uddin, S. Mansor, and M. G. Kibria, "SCCM: An Interpretable Enhanced Transfer Learning Model for Improved Skin Cancer Classification", *BioMedInformatics*, vol. 5, p. 43, 2025, doi: [10.3390/biomedinformatics5030043](https://doi.org/10.3390/biomedinformatics5030043).

Conference Papers

- F. Rabby, R. Das, M. M. Rahman, M. H. Hossain, and **M. R. Aknda**, "Scalable Hand Gesture Recognition from Surface Electromyography (sEMG) Signals Using a Hybrid Deep Learning Model Evaluated on Diverse Datasets", *2025 9th International Conference on Electrical, Electronics and Information Engineering (ICEEIE)*, Mataram, Indonesia, 2025, pp. 1–6, doi: [10.1109/ICEEIE66203.2025.11252161](https://doi.org/10.1109/ICEEIE66203.2025.11252161). (**Best Paper Award**)
- F. Rabby, **M. R. Aknda**, M. T. Mahi, S. R. Ahmed Ratul, and B. Z. Shezan, "TinySenseNet: A Lightweight sEMG-IMU Fusion Network Using TinyML for

Mechanical Arm Control in Low-Resource Settings,” 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS), Kushtia, Bangladesh, 2025, pp. 1–6, doi: [10.1109/COMPAS67506.2025.11381831](https://doi.org/10.1109/COMPAS67506.2025.11381831).

- F. Rabby, M. H. Hossain, **M. Rifat Aknda**, B. Z. Shezan, and S. Salwa Mou, “Real-Time Brain Tumor Localization and Classification Using YOLOv11 and MONAI MedNeXt,” 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS), Kushtia, Bangladesh, 2025, pp. 1–6, doi: [10.1109/COMPAS67506.2025.11381706](https://doi.org/10.1109/COMPAS67506.2025.11381706).

Research Projects

SCCM: Interpretable Transfer Learning Model for Skin Cancer Classification (2024–2025)

- Designed and evaluated CNN-based transfer learning architectures for dermoscopic image classification.
- Fine-tuned pretrained backbones and assessed performance using internal and external validation datasets.
- Applied Grad-CAM++ for explainable AI-based clinical interpretation and systematic error analysis.
- Achieved 90.91% test accuracy and 93.73% external validation accuracy.
- Published findings in a peer-reviewed journal.

Multimodal sEMG-IMU Based Gesture Recognition (2024–2025)

- Developed hybrid deep learning architectures for gesture recognition using surface electromyography (sEMG) and Inertial Measurement Units (IMUs) signals.
- Explored signal preprocessing and representation learning for time-series biomedical data.
- Performed cross-dataset evaluation to analyze robustness and scalability.
- Contributed to feature engineering, model training, and experimental result analysis.
- Investigated implications for real-time human-computer interaction using wearable sensor data.
- Published findings in peer-reviewed conference papers.

Technical Skills

Programming: Python, R, C/C++

Machine Learning: Deep Learning, CNNs, Transformers, Transfer Learning, Multimodal Machine Learning, Feature Fusion, LLMs, RAG, Agentic AI

Medical AI: Medical Imaging, Explainable AI (Grad-CAM++, SHAP)

Frameworks: PyTorch, TensorFlow, scikit-learn, MONAI

Research Tools: LaTeX, Git, Jupyter Notebook

Research & Industry Experience

AI Developer | Softvence (Betopia Group)

February 2026 – Present

- Develop and fine-tune AI and large language models (LLMs) for real-world applications.
- Implement Generative AI, RAG pipelines, and agentic AI for autonomous reasoning and knowledge retrieval.
- Collaborate with internal teams and clients to design, develop, and evaluate real-world AI solutions.

AI Engineer | Research & Development Dept. | EVOMED Technology

April 2025 – September 2025

- Conducted data preprocessing, exploratory data analysis, and evaluation of user-facing healthcare machine learning systems.
- Supported experimentation and validation of deep learning models on real-world medical datasets.
- Collaborated with other researchers and multidisciplinary teams to document experimental results and system behavior.

Awards & Academic Activities

- **Best Paper Award**, ICEEIE (2025)
- Vice President, ULAB Computer Programming Club (UCPC) (2023)

Certifications

- Digital Security Training (DST) - Training of Trainers (ToT)
Asia Centre & Google, In collaboration with Dept. of CSE, ULAB (Nov 2025)
- Big Data Fundamentals with PySpark | DataCamp (Jan 2025)

Academic Referees

Prof. Muhammad Golam Kibria, PhD, SMIEEE

Professor & Head, Department of Computer Science and Engineering, University of Liberal Arts Bangladesh (ULAB)

Supervisor - Capstone Project & Internship

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